UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

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|---|---|---|
| COURSE | : | PRE CALCULUS |
| COURSE CODE | : | MAT133 |
| EXAMINATION | : | NOVEMBER 2026 |
| TIME | : | 2 HOURS |

MARKING SCHEME NOTES

1. This marking scheme accompanies the MAT133 mock paper (November 2026).
   Answers are unofficial model answers generated from the printed past-paper question; they are not an official UTM marking scheme.

2. Marking points follow the printed part labels and marks. Verify every model answer against the source page.

3. Four questions: 10 + 10 + 10 + 25 marks; total = 55 marks.

4. Questions: 4. Real recycled questions: 4. Placeholders: 0.
   For local study use only; model answers are unofficial.

5. Every question is a real MAT133 past-paper question; its source paper and page are printed with the answer.

FOR MARKERS' USE ONLY

*This marking scheme consists of 4 printed pages*

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## QUESTION 1 (10 marks)

Source: MAT133-2025-JUL, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Given that $f(x) = 2x^2 + 3$ and $g(x) = \sqrt{x+2}$.

i) State the domain of $f(x)$ and $g(x)$.
(2 marks)

ii) Determine $(f \circ g)(x)$.
(3 marks)

Answer: Domain of $f$: $(-\infty, \infty)$; domain of $g$: $[-2, \infty)$ (i.e. $x \geq -2$). | $(f \circ g)(x) = 2x + 7$ for $x \geq -2$.

- $f(x) = 2x^2 + 3$ is a polynomial, so its domain is all real numbers.
- $g(x) = \sqrt{x+2}$ requires $x + 2 \geq 0$, so $x \geq -2$.
- $(f \circ g)(x) = f(g(x)) = 2(\sqrt{x+2})^2 + 3$
- $= 2(x+2) + 3 = 2x + 7$
- Domain restriction from $g$: $x \geq -2$.

b) Given that $m(x) = \frac{2x+3}{2x-3}$, find the value of x if $m^{-1}(x) = x$.
(5 marks)

Answer: $x = -\frac{1}{2}$ or $x = 3$

- Let $y = m^{-1}(x)$. Then $x = m(y) = \frac{2y+3}{2y-3}$.
- Solve for $y$: $x(2y-3) = 2y+3 \to 2y(x-1) = 3x+3 \to y = \frac{3(x+1)}{2(x-1)}$, $x \neq 1$.
- Set $m^{-1}(x) = x$: $\frac{3(x+1)}{2(x-1)} = x$.
- $3x + 3 = 2x^2 - 2x \to 2x^2 - 5x - 3 = 0$.
- $(2x+1)(x-3) = 0 \to x = -\frac{1}{2}, 3$. Both satisfy $x \neq 1$.

Assumptions: For part (a)(ii), the domain of the composite function is taken as the domain of $g$, namely $x \geq -2$. No figure is required.

Model confidence: high.

## QUESTION 2 (10 marks)

Source: MAT133-2021-JUL, page(s) 7

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Solve the inequality $\frac{2}{3+2x} \leq -2$. Express the answer in interval notation.
(5 marks)

Answer: $\left(-\frac{3}{2}, -2\right]$

- $\frac{2}{3+2x} + 2 \leq 0 \to \frac{8+4x}{3+2x} \leq 0$
- $\frac{4(x+2)}{3+2x} \leq 0$
- Critical values: $x = -2$ (included), $x = -\frac{3}{2}$ (excluded).
- Sign: positive on $(-\infty, -\frac{3}{2})$, negative on $(-\frac{3}{2}, -2)$, positive on $(-2, \infty)$.
- Solution: $(-\frac{3}{2}, -2]$.

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b) If $Z_1 = 1 - 2\mathrm{i}$ and $Z_2 = -3 + 5\mathrm{i}$, state $\frac{\overline{Z}_2 \mathrm{i}^{31}}{Z_1}$ in standard form $a + bi$.
(5 marks)

Answer: $-\frac{11}{5} - \frac{7}{5}i$

- $\overline{Z_2} = -3 - 5\mathrm{i}$
- $\mathrm{i}^{31} = \mathrm{i}^{28}\mathrm{i}^3 = -\mathrm{i}$
- $\overline{Z_2}\,\mathrm{i}^{31} = (-3 - 5\mathrm{i})(-\mathrm{i}) = -5 + 3\mathrm{i}$
- $\frac{-5+3\mathrm{i}}{1-2\mathrm{i}} \cdot \frac{1+2\mathrm{i}}{1+2\mathrm{i}} = \frac{-11-7\mathrm{i}}{5}$
- $= -\frac{11}{5} - \frac{7}{5}\mathrm{i}$

Model confidence: high.

**QUESTION 3** **(10 marks)**

Source: MAT133-2022-FEB, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Given two points $A(-4, p)$ and $B(2, 7)$ located on the equation $y = 2x + 3$. Find the value of $p$. Hence, find the equation of a straight line which is perpendicular to the line $AB$ and passes through point $A$.
(5 marks)

Answer: $p = -5$. The perpendicular line through $A(-4, -5)$ is $y = -\frac{1}{2}x - 7$, or equivalently $x + 2y + 14 = 0$.

• Since $A(-4, p)$ lies on $y = 2x + 3$, $p = 2(-4) + 3 = -8 + 3 = -5$.

• Thus $A(-4, -5)$ and line $AB$ has slope $2$ because it is $y = 2x + 3$.

• Perpendicular slope $m = -\frac{1}{2}$.

• Using point-slope through $A$: $y + 5 = -\frac{1}{2}(x + 4)$.

• Simplify: $y = -\frac{1}{2}x - 7$, i.e. $x + 2y + 14 = 0$.

b) Find the standard equation of circle where the diameter has endpoints $(-1, -8)$ and $(5, 0)$. Hence sketch the graph.
(5 marks)

Answer: Standard equation: $(x-2)^2 + (y+4)^2 = 25$. Sketch: circle with centre $(2, -4)$ and radius $5$, passing through the diameter endpoints $(-1, -8)$ and $(5, 0)$; $x$-intercepts at $(-1, 0)$ and $(5, 0)$; $y$-intercepts at $(0, -4 \pm \sqrt{21})$.

• Centre $C = \left(\frac{-1+5}{2}, \frac{-8+0}{2}\right) = (2, -4)$.

• Diameter length $= \sqrt{(5-(-1))^2 + (0-(-8))^2} = \sqrt{6^2 + 8^2} = 10$, so radius $r = 5$.

• Standard form: $(x-h)^2 + (y-k)^2 = r^2$ with $(h, k) = (2, -4)$, $r = 5$.

• Thus $(x-2)^2 + (y+4)^2 = 25$.

• Sketch as a circle centred at $(2, -4)$ with radius $5$; label endpoints $(-1, -8)$ and $(5, 0)$.

Assumptions: The sketch in part b is described by its centre, radius and labelled diameter endpoints because JSON cannot contain a drawn graph.

Model confidence: high.

**QUESTION 4** **(25 marks)**

Source: MAT133-2023-JUL, page(s) 3

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Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Given that $f(x) = x + 7$ and $g(x) = \frac{x+3}{x-4}$.

i) State the domain of $f(x)$ and $g(x)$
(2 marks)

ii) Determine $(g \circ f)(x)$. Hence, find $(g \circ f)(4)$.
(3 marks)

iii) Find the value of $x$ if $(g \circ f)^{-1}(x) = 2$
(4 marks)

Answer: i) Domain of $f$: $x \in \mathbb{R}$; domain of $g$: $x \in \mathbb{R},\ x \neq 4$. ii) $(g \circ f)(x) = \frac{x+10}{x+3}$ and $(g \circ f)(4) = 2$. iii) $x = \frac{12}{5}$.

• f(x)=x+7 is defined for all real x.
• g(x)=(x+3)/(x-4); denominator cannot be 0, so x-4≠0, i.e. x≠4.
• $(g \circ f)(x) = g(x+7) = \frac{(x+7)+3}{(x+7)-4} = \frac{x+10}{x+3}$.
• Need x+7≠4, so x≠ -3.
• $(g \circ f)(4) = \frac{4+10}{4+3} = \frac{14}{7} = 2.$
• Let h=g° f. h^-1(x)=2 ⇒ x=h(2).
• $h(2) = \frac{2+10}{2+3} = \frac{12}{5}.$
• Thus x=(12)/(5).

b) Use synthetic division to show that $(x-1)$ is not a factor of $f(x) = 2x^3 - x^2 + 1$. Hence, state the quotient and remainder of $f(x)$.
(5 marks)

Answer: Quotient: 2x^2+x+1; Remainder: 2; hence (x-1) is not a factor.
• Synthetic division with c=1; coefficients of f(x)=2x^3-x^2+0x+1 are 2,-1,0,1.
• Bring down 2; multiply by 1 and add: 2(1)+(-1)=1; then 1(1)+0=1; then 1(1)+1=2.
• Quotient is 2x^2+x+1 and remainder is 2≠0.

c) Find the value of $x$ if $\frac{3^x - 9(3^{-x})}{2} = 4$.
(6 marks)

Answer: x=2
• $\frac{3^x - 9(3^{-x})}{2} = 4 \rightarrow 3^x - 9(3^{-x}) = 8.$
• Let y=3^x>0. Then y-(9)/(y)=8 ⇒ y^2-8y-9=0.
• (y-9)(y+1)=0. Since y>0, y=9.
• $3^x = 9 = 3^2 \rightarrow x = 2.$

d) Based on the graph of $y = x^2$, sketch the graph of $y = -[(x+2)^2 - 1]$ by using transformation technique. Hence, state the $x$ and $y$ -intercept(s) (if any).
(5 marks)

Answer: y=1-(x+2)^2; vertex (-2,1), opens downward; x-intercepts: (-3,0),(-1,0); y-intercept: (0,-3).
• Rewrite y=-[(x+2)^2-1]=-(x+2)^2+1.
• From y=x^2: shift left 2, reflect in the x-axis, then shift up 1.
• x-intercepts: set y=0 ⇒ (x+2)^2=1 ⇒ x=-3,-1.
• $y - intercept : set x = 0 \rightarrow y = -3.$

Assumptions: For part (d), the sketch is described algebraically from the given equation; no additional graph features are assumed.

Model confidence: high.

END OF MARKING SCHEME

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